

## **Personal Statement**

### **About me**

We all have fears. To me, fear means a choice: Forget Everything And Run, or Face Everything And Rise. I choose the latter. My biggest fear isn't dramatic; it's the quiet fear of staying small in mindset, ambition, and action. Growing up in a big city in Vietnam, I was exposed to technology early. Seeing a real robot at my school club sparked my passion for tech and showed me I could build things that matter. Since then, I've spent years nurturing that passion, from high school to university. Classes, projects, and competitions have steadily expanded my knowledge. I've failed quietly more times than I can count, but I've learned even more. And the deeper I dive into building technology, the more I understand: it's not just about algorithms; it's about application. With that motivation, I now continue my studies in Singapore, where I am currently a penultimate-year Computer Science undergraduate at National University of Singapore. My primary focus areas are software engineering and artificial intelligence, as I love building delightful software products that bring AI ideas to practical use and make a real difference.

### **Working experience**

To broaden my experience in the tech industry, my first role was as a software engineering intern at a Singapore-based EdTech startup focused on technical interview preparation. There, I contributed to maintaining and revamping the company's main educational platform, which serves an average of 500 users daily. In addition to my technical responsibilities, I collaborated with the marketing team to develop technical marketing content aimed at increasing brand awareness. This experience challenged me to adapt quickly to new tasks, such as technical marketing writing, which were initially unfamiliar to me. With guidance from the marketing lead, I successfully produced high-quality content that helped the company grow its follower base by 3,000 in four months. This experience not only strengthened my adaptability but also highlighted the value of interdisciplinary teamwork.

After my first internship, I joined a research lab where I saw how innovative AI ideas can solve real-world problems. There, I worked on developing an autonomous navigation system for a low-cost robot, applying various AI techniques. Although the project was initially challenging, it was fascinating to put my classroom knowledge of machine learning and artificial intelligence into practice. The experience was highly rewarding, as I explored new concepts through research papers and contributed to coding the navigation system. As a result, my contribution to writing an optimal path selection algorithm for the navigation framework helped the team clinch the first prize, scoring 79% of the total possible points at The EarthRover Challenge 2025.

### **How can I contribute**

With experience in both startups and research labs, I have learned to adapt quickly and solve problems in different situations. I am not afraid to step out of my comfort zone, and I always work hard to learn new things and get better at what I do. I enjoy turning complicated technical ideas into useful solutions that make a real difference for people. I also work well with others, whether it's engineers, marketers, or researchers, and I know how to explain technical ideas in simple ways. My background in software engineering and artificial intelligence helps me build strong, delightful products and add helpful AI features. I am eager to help by building reliable software, using AI to solve real problems, working with different teams, and making sure everyone understands our work. I believe my positive attitude, technical skills, and teamwork will help me contribute from the very first day and support the team's success.

### **Internship objectives**

With my new journey during the NUS Overseas College Programme, I would like to immerse myself in working in a startup environment different from Singapore. I want to work alongside passionate entrepreneurs not just to contribute technically, but to witness how ideas become products, and how products scale to create change. Through this experience, I hope to gain hands-on exposure in a dynamic international startup ecosystem, develop an enterprising mindset by learning directly from founders and mentors, and expand my global network. By combining practical work with academic insights, I aim to deepen my understanding of building and growing impactful ventures, equipping myself with the skills and perspectives needed to drive innovation and positive change in the future.

Thank you for considering my application, and I look forward to your acceptance.

## Education

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|---------------------|------------------------------------------------------------------------------------------------|-----------|
| Aug 2023 - Present  | <b>National University of Singapore</b><br>Bachelor of Computing (Honours) in Computer Science | Singapore |
| Sep 2020 – May 2023 | <b>Le Hong Phong High School for The Gifted</b><br>High School Diploma                         | Vietnam   |

## Work Experience

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| May 2025 – July 2025 | <b>Amaris.AI</b><br><i>AI Engineering Intern</i> <ul style="list-style-type: none"><li>Implemented company AI research tool's automated report generation pipelines that aggregate, analyze, and summarize search results from multiple sources, streamlining knowledge extraction and delivery for end-users.</li><li>Developed and optimized API key management and rotation logic, ensuring high availability and uninterrupted access to third-party search/data services.</li><li>Designed and implemented an Agent-2-Agent backend system enabling secure, efficient, and automated data exchange between multiple services.</li></ul>       | Singapore |
| Mar 2025 - May 2025  | <b>Smart Systems Institute</b><br><i>Researcher and Developer Intern</i> <ul style="list-style-type: none"><li>Manually annotated road images to fine-tune a pre-trained SAM model on a dataset of over 15,000 images, achieving segmentation accuracy above 90%.</li><li>Utilized a state-of-art calibration method to calibrate robot's camera, significantly reducing error in converting between world coordinates and image coordinates.</li><li>Designed an effective cost function and applied k-means algorithm to select most optimal candidate trajectory for robot, significantly eliminating unnecessary or dangerous paths.</li></ul> | Singapore |
| Sep 2024 – Jan 2025  | <b>GreatFrontEnd</b><br><i>Software Engineering Intern</i> <ul style="list-style-type: none"><li>Developed key features and resolved critical technical bugs in an EdTech startup's coding interview platform, enhancing stability and performance with Next.js and TailwindCSS.</li><li>Collaborated with marketing team to design technical and promotional content, contributing to a growth of 3,000 subscribers across company's social media accounts.</li><li>Verified and authored detailed solution explanations for 100+ data structures and algorithms (DSA) questions, improving user learning experience.</li></ul>                   | Singapore |

## Projects

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| Jul 2024 - Sep 2024 | <b>Gestura</b> <ul style="list-style-type: none"><li>Developed an AI-powered social application using Next.js to detect sign languages, resulting in a user experience enhanced for deaf-mute community.</li><li>Finetuned MediaPipe gesture recognizer model to accurately detect and interpret sign language gestures in videos, resulting in an accuracy of 40%.</li><li>Developed AI-powered backend system using FastAPI, creating restful API to handle video uploads, processing, and AI model's detection result retrieval.</li></ul> | Singapore |
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| May 2024 - Jul 2024 | <b>Foodie</b> <ul style="list-style-type: none"> <li>Implemented robust backend functionality with PostgreSQL, and Go, including user authentication, post management, follow requests, and file uploads utilizing restful API and JWT (JSON Web Tokens) for secure authentication.</li> <li>Designed and structured backend architecture with modular components such as controllers, models, routes, and middleware for efficient development and maintenance.</li> <li>Utilized Pusher's Web Sockets to facilitate instant message delivery for effective real-time communication between multiple users.</li> </ul> | Singapore |
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## Scholastic Achievements/Extracurricular Activities

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| May 2025            | <b>Earth Rover Challenge (ERC) at IEEE International Conference on Robotics &amp; Automation 2025</b> <ul style="list-style-type: none"> <li>First Prize</li> </ul>   | Singapore |
| Jan 2024 - May 2024 | <b>School Contributions</b> <ul style="list-style-type: none"> <li>Teaching Assistant for CS1010 Programming Methodology course at NUS School of Computing</li> </ul> | Singapore |
| March 2023          | <b>Ho Chi Minh City Science and Engineering Fair</b> <ul style="list-style-type: none"> <li>First Place in the Intelligent Systems Category</li> </ul>                | Singapore |

## Skill Sets & Proficiency

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|---------------------------------------|----------------------------------------------------------|
| <b>Programming</b>                    | Python<br>Go<br>Javascript<br>Typescript<br>SQL          |
| <b>Web</b>                            | HTML, CSS                                                |
| <b>Database</b>                       | PostgreSQL<br>Google Firebase                            |
| <b>Web Frameworks and Libraries</b>   | FastAPI<br>React<br>Next.js<br>Tailwind CSS              |
| <b>AI/ML Frameworks and Libraries</b> | Pytorch<br>Numpy<br>Pandas<br>Matplotlib                 |
| <b>DevOps Tools</b>                   | Docker                                                   |
| <b>Tools</b>                          | Git<br>Github                                            |
| <b>Office Productivity</b>            | Microsoft Word, PowerPoint<br>Microsoft Excel            |
| <b>Non-technical Skills</b>           | Outlining technical content blogs/posts<br>Video editing |

## Language Proficiency

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| <b>Spoken</b>  | English – fluent; |
| <b>Written</b> | English – fluent; |